Microdata User Guide

Canadian Internet Use Survey

2022

Statistics Canada Statistique Canada

Canada

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1. Introduction

The Canadian Internet Use Survey (CIUS) was conducted by the Center for Innovation, Technology and Enterprise Statistics of Statistics Canada on behalf of Innovation, Science and Economic Development Canada (ISED) in 2022. This manual has been produced to facilitate the manipulation of the microdata file of the survey results.

Any question about the data set or its use should be directed to:

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2. Background

The Household Internet Use Survey (HIUS) was first conducted in 1997, and ran annually until 2003. The HIUS focused on household Internet penetration. In 2005, the CIUS replaced the HIUS. The redesign in 2005 focused more on Internet use by individuals, while conforming to international standards regarding statistical indicators for Internet access and use. From 2005 to 2009, the CIUS was conducted biennially.

In 2010, the CIUS was redesigned to meet the measurement needs of Broadband Canada: Connecting Rural Canadians Program sponsored by Industry Canada (now ISED). As a hybrid survey on access and use, the CIUS measured the type, speed and cost of household Internet access and the individual online behaviours of a selected household member.

Redesign in 2018:

While previous survey iterations were administered as a supplement to the Labour Force Survey (LFS), the 2018 CIUS was redesigned and became a standalone survey. It focussed on measuring individual's use of digital technologies, the Internet, and online behaviours including; those related to social media, e-commerce, online government services, online work, digital skills, streaming content and security privacy and trust as it relates to the Internet. The survey also measures the access to internet as well as the type and speed of the connection at home.

Several changes related to the methodology of the survey (frame, collection method, etc) in 2018, involved that the 2018 CIUS findings should not be compared with those from previous surveys.

In order to take account of the rapid evolution of Internet-related technologies, the 2020 and 2022 cycles have been updated to include new content. To make room for this content without adding to the response burden of respondents, some questions or modules were rotated out. Please note that there may be minor changes in the question wording, notes, inclusions and exclusions between cycles. Refer to the questionnaires when making comparisons between cycles.

The CIUS collects various demographic characteristics from the respondent through the questionnaire such as age, gender, household relationships, level of education, indigenous identity, and location (postal code), etc.

As of 2020, immigration status and family income data are obtained through data linkage to administrative data from the Longitudinal Immigration Database (IMDB) and the Annual Income Estimates for Census Families and Individuals (T1 Family File) T1FF tax data. In 2022, the language questions were added to the list of linked variables, and were obtained using the 2021 Census.

3. Objectives

The 2022 CIUS aims to measure the adoption and use of digital technologies by individual residents of Canada 15 years of age and over, living in the provinces. The information gathered will help to better understand how the Internet and other digital technologies are changing the way we work, play and interact with others.

The CIUS examines Internet access and use, the use of Internet-connected smart devices and smartphones, social connections in the digital age, the use of government online services, e-commerce, digital skills, online security, privacy and trust, online work and the knowledge and adoption of new digital technologies such as Artificial Intelligence, digital credentials and cryptocurrencies. The CIUS also measures barriers to: Internet access and use, online services, and using various digital technologies.

Collected data is used to inform evidence-based policymaking, research, and program development, and provide internationally comparable statistics on the use of digital technologies. For example, the data of this survey is used to:
- Guide government efforts to provide households with more reliable and affordable high-speed Internet
- Develop policies to protect individuals from online privacy and security risks
- Research the impacts of digital technologies on well-being and new gig-based employment
- Better understand the digital skills needed for learning and the future of work
- Better understand how and why people use online services, like shopping and social media,
- Identify barriers that prevent people from accessing the Internet and making the most of new technology
- Improve online government services and make them more user-friendly
- Contribute to international initiatives, such as the United Nations Sustainable Development Goals and the OECD Going Digital Project, to help track and compare Canada's digital development.

The survey is built off the previous iterations of the CIUS, last conducted in 2018 and 2020. The 2022 iteration has been updated to collect data to meet new data needs.

The 2022 iteration of the CIUS is sponsored by Innovation, Science and Economic Development Canada (ISED). Numerous other government departments also provided input during the questionnaire content development phase.

4. Concepts and Definitions

4.1 Canadian Internet Use Survey Concepts and Definitions

Absence from work	A period of three months or more when a person stops working for reasons such as lack of work, sickness, maternity/paternity leave, retirement, or any other reason. This could be paid or unpaid.
App	A software that can run on the Internet, mobile devices or computers. They are downloaded by a user and are designed to perform functions, activities or tasks to benefit the user.
Artificial Intelligence	Artificial Intelligence, or AI, is the set of computer techniques that enable a machine (e.g., a computer or mobile telephone) to perform tasks normally requiring human intelligence, such as reasoning or learning.
Automatic updates	Updates that are done automatically via the Internet, not requiring manual programming or periodic authorization.
Biometric security features	Measures unique characteristics of a person, such as voice pattern, the iris or retina pattern of the eye, or fingerprint patterns, to authenticate a user and provide access to personal accounts.
Blog	A blog (short for 'weblog') is a public website where users post informal journals of their thoughts, comments and philosophies, often updated frequently and normally reflecting the views of the blog's creator. Blogs provide commentary or news on a particular subject such as food, politics or local news; some function as personal online diaries. The term 'blogging' is used as an example of contributing content online.
Bluetooth	A standard for the short-range wireless interconnection of mobile phones, computers, and other electronic devices.
Broadband	Refers to telecommunication in which a wide band of frequencies is available to transmit information. Because a wide band of frequencies is available, information can be sent on many different frequencies or channels within the band at the same time, allowing more information to be transmitted in a given amount of time (much as more lanes on a highway allow more cars to travel on it at the same time). Broadband internet connections allow connection to the World Wide Web digitally, rather than through use of the dial-up service. A digital connection (DSL and cable) is faster and allows usage of a traditional phone line while simultaneously being connected to the Internet.
Cable Internet	Many Internet Service Providers (e.g., Rogers, Vidéotron, Shaw, Teksavvy) offer Internet services by cable. Coaxial cable that was traditionally used to provide television services can be upgraded to also facilitate Internet access. Over time, Internet by cable has become available in many (though not all) areas where cable television is available.

Child(ren)	**Adopted:** Child(ren) for whom the respondent signed legal documents giving him/her the authority to raise the child(ren) as his/her own. **Birth:** Refers to biological sons and daughters. **Stepchildren:** Children from a former union of a current or former spouse or common-law partner. **In shared or joint custody:** Those who spend time with each parent and who spent the previous night (in reference to the day of the interview) at the other parent's residence.
Common-law partner	Refers to two people of the opposite sex or the same sex who live together as a couple but who are not legally married to each other.
Co-working spaces	Coworking is a way of working that includes the use of an office or other working environment by people who are self-employed or working for different employers, typically so as to share office space, equipment, ideas, and knowledge.
Created presentations, or documents with text and pictures, tables, or charts	This activity includes, but is not limited to, the use of computer software specifically designed for creating presentations, like Microsoft Powerpoint, Prezi and Keynote. The activity of creating presentations can be conducted in programs not specifically designed for the development of presentations, but should contain multiple elements, such as text with pictures, text with tables, or text with charts.
Crowd-based microwork	The collaboration of individuals and businesses to complete one-off jobs in exchange for payment. Usually, this type of work over the Internet is secured by individuals posting their skills on an online job board which businesses can access, or businesses post their need for work to be completed on a job board, at which point accessing individuals can apply for the work. Crowd-based microwork jobs could include describing a product, finding duplicate webpages or writing algorithms.
Cryptocurrency	A digital currency in which encryption techniques are used to regulate the generation of units of currency and verify the transfer of funds, operating independently of a central bank.
Cyber-ransom	Cyber-ransom is the exercise of extortion over the Internet, whereby someone prevents users of technologies from interacting with their files, applications or systems until a payment is made. **Ransomware** is the term used to describe the malicious software designed to block users' access.
Cyber security software	Also known as cybersecurity software, any computer program designed to enhance information security and defend computer devices against intrusion and unauthorized use of resources.
Data storage service	A service used for archiving data in forms for use by a computer or other electronic device.
Desktop computer	A personal computer (PC) intended for regular use at a single location, as opposed to a mobile laptop or portable computer.

Dial-up through telephone line	Dial-up is a method of connecting to the Internet using a traditional telephone line. A modem is required, and the bandwidth (or transmission speed) is limited to 64 kilobits per second (kbps).
Download speed	Refers to how fast a computer or other electronic device obtains data from a remote location, either through cables or a type of wireless connection.
Digital subscriber line (DSL)	Refers to high-speed Internet access over telephone lines. The service is many times faster than traditional dial-up Internet access. The technology consists of twisted copper pair that has been upgraded to offer the capacity to support Internet service. DSL may also be known as ADSL (Asymmetric DSL), and a fast version of the service is sometimes known as VDSL (Very-high-bitrate DSL). Another main difference between high-speed DSL packages and traditional dial-up services is that users' telephone lines remain available to receive or make telephone calls while connected to the Internet using high-speed DSL connections. (With dial-up Internet service, the telephone line is typically unavailable for phone calls, except in special cases where users pay a fee in order to have their dial-up Internet connection interrupted in order to receive calls whenever a telephone call comes in).
Dwelling	A dwelling is a set of living quarters in which a person or a group of people reside or could reside (e.g. a private home, an apartment, a duplex).
E-book	Also known as an electronic book (audio book or podcast book), this book publication is in digital form, consisting of text, images or both, readable on a flat-panel display of computers or other electronic devices.
E-book reader	Also known as an electronic book, this book publication is in digital form, consisting of text, images or both, readable on a flat-panel display of computers or other electronic devices.
Elected official of local government	A local or municipal government is responsible for services such as firefighting, maintaining city streets and other local matters. Local or municipal governments must follow laws and guidelines as set out by their province but are allowed to pass additional by-laws and acts unique to them. Municipalities may also be part of a larger county or regional government, which are also considered to be local governments. Common names for local governments include region, county, district, township, town, borough, village, etc. Examples of people who are elected to local or municipal governments include Mayors and Councillors. If there is no local or municipal government, services are provided by the provincial or territorial government.
Electronic bank transfer	A method of payment over the Internet, whereby funds are transferred between bank accounts. It may take the form of an Interac email money transfer. To deposit money received in an email money transfer, users are typically required to answer a security question and to have their bank account set up for online banking.
Enabled automatic updates for, or manually updated, operating systems	Most security software provides an option to users to download and install updates automatically at regular intervals, or to update manually. Security updates may contain new virus definitions or updates to protect against new or emerging security threats. Updates do not refer to subscription renewals, such as annual or

on any of your devices	monthly fees required using the software.
eSports	eSports are multiplayer video games that are typically played by professional gamers competitively for spectators.
Federal Member of Parliament	A Federal Member of Parliament, including the Prime Minister: The federal government is responsible for things such as citizenship and immigration, national defence, and trade with other countries. Members that are elected to the House of Commons are called Members of Parliament (MPs) and usually belong to a political party. The Prime Minister is the head of the federal government.
Fiber-optic	A fiber-optic cable is a network cable that contains strands of glass fibers inside an insulated casing. It is designed for long distance, very high-performance data networking and telecommunications. Compared to wired cables, fiber optic cables provide higher bandwidth and can transmit data over longer distances.
Fixed wireless, point-to-point	A fixed wireless, point-to-point connection (requiring line of site reception) is a form of wireless Internet access and is found mostly in rural areas. It works by sending an Internet signal from a central distribution tower to a receiver, which may be located on the roof of a dwelling. In order for this service to work, homes receiving the signal must have clear 'line-of-sight' reception from the transmission tower. That is, the technology requires a path unobstructed by hills or large trees in the vicinity.
Foster father or mother	A person who is paid to provide care for a child less than 18 years old, and who is not the birth, adoptive or step parent of the child.
Foster son or daughter	A child less than 18 years old who is receiving care from someone who is being paid to provide that care, and who is not the child’s birth, adoptive or step parent.
Fraudulent content	Emails or messages sent from social media sent to obtain personal or financial information from recipients and gain financially by being intentionally deceptive.
Fraudulent emails	Fraudulent emails are emails sent to obtain personal or financial information from recipients and gain financially by being intentionally deceptive.
Full-time employment	Respondents who work for pay or in self-employment 30 or more hours per week.
Gaming console	A gaming console is a specialized computer system designed for interactive video gameplay and display. A video game console functions like a personal computer (PC) and is built with the same essential components. Xbox, PlayStation and Nintendo are examples of video gaming consoles.
Getting redirected to fraudulent websites	Arriving involuntarily at websites that are claiming or appearing to have been set up by the legitimate provider.

Gig economy	Workers in the gig economy are usually not employed on a long-term basis by a single firm; instead, they enter into various contracts with firms or individuals (task requesters) to complete a specific task or to work for a specific period of time, either in person or digitally, for which they are paid a negotiated sum. Examples include Uber, Lyft, TaskRabbit, Upwork, Guru, Fiverr, Freelancer, or incomes earned through online advertisements and sponsored content.
Household	Refers to a person or a group of persons (other than foreign residents) who occupy the same dwelling and do not have a usual place of residence elsewhere in Canada. It may consist of a family, of a group of unrelated persons (e.g. roommates), or of one person living alone.
Household member	A household member is any person who, during the survey period: • regards or is reported to regard the dwelling as their usual place of residence; or • is staying in the dwelling and has no other usual place of residence elsewhere. Household members who are temporarily absent on the day of the survey are considered as part of the usual household.
Husband or wife	Spouse by legal marriage (same sex or opposite sex).
Indigenous	Indigenous people are usually those with ancestors who resided in North America prior to European contact and who identify with one of the three Indigenous groups listed: First Nations (North American Indian), Métis and Inuk.
In-law	A person who is related by marriage only.
Institution	A nursing home, hospital, convalescent home, rest home, penal institution, or other long-term care facility.
Instant messaging app	An instant messaging app is a common form of online communication where two or more persons exchange text to simulate a conversation. In addition to exchanging text, many instant messaging programs also allow users to share photos, or to display their Webcam. Services such as Facebook Messenger or Google Chat are other examples of instant messaging programs. BBM and iMessage are instant messaging apps because they use Internet-based infrastructure.
Junk mail	Junk mail is unwanted or unsolicited advertising or promotional material received through email.
Laptop or netbook	Laptop or netbook is a portable computer, usually battery-powered, small enough to rest on the user's lap and having a screen that closes over the keyboard like a lid. A netbook is a small, lightweight version of a laptop with many of the same computing features (e.g., surf the web, run software).
Loyalty programs	Loyalty programs are structured or long-term marketing efforts by companies which provide incentives to repeat customers demonstrating loyal behaviour. Often, following registration to the program, customers may accumulate points for

	transactions that can be used to make future purchases. Examples of paid online loyalty programs include Amazon Prime, Plum Plus, PC Insiders etc.
Loyalty program points fraud	Loyalty program points fraud is an act of wrongful or criminal deception to unjustly claim, steal or redeem points collected by another person.
Media streaming device	Is what connects your television or home theater to the Internet and allows you to stream video and music from online services. Examples include Apple TV, Amazon Fire TV, Google Chromecast.
Misinformation	Misinformation is information which was shared or spread that is false, regardless of intent.
Mobile data plan	A data plan is a mobile Internet service for a wireless handheld device that allows users to access the Internet from anywhere via a 3G, 4G, 5G or LTE network. The service is provided by mobile operators for a fee that may be billed monthly or on a per-usage basis.
Municipal wireless	Municipal wireless is a citywide wireless network, which is usually achieved by providing municipal broadband via Wi-Fi to large parts, or all of a municipal area. The typical deployment design uses hundreds of wireless access points positioned outdoors, often on poles. The operator of the network acts as a wireless Internet service provider.
Mobile hotspots	Mobile hotspots are a recent technology that allows you to connect your Internet capable devices to the Internet through a wireless, portable device. The hotspot forms an on-the-spot Wi-Fi network and you can connect a number of computers or gadgets to the network for simple, fast Internet access. Note: LTE is a standard for high speed wireless communication for mobile devices and data terminals.
Online bulletin board	An online bulletin board is an online version of a typical market, whereby products or services for sale are displayed for purchase or hire by individuals. The payment for goods or services obtained or sold through online bulletin boards is done in person, at the time of the sale
Online freelancing	Online freelancing is a paid work secured and submitted to the contracting party over the Internet by someone working by the hour, day, or job, rather than on a part-time or full-time regular or salaried basis. Some examples of online freelancing are data entry, translation, photo editing or website design.
Online platform	An online platform is an online marketplace that places one party in touch with another, such as buyers and sellers. Examples are eBay, Craigslist, Amazon Marketplace, Airbnb and Uber. The online system may be entirely self-controlled or it may allow third-party apps to connect via the platform's programming interface (API).
Online data storage space	Refers to the storing of electronic data, such as files or photos to a third party service via the Internet. This could be a paid or free service.
Operating system	An operating system (OS) is the software that supports a computer or other

(OS)	electronic device's basic functions, through which applications for varied uses can interact and be installed.
Organizations	Organizations or agencies may be government or privately run and are staffed with paid individuals or volunteers who provide care. Some examples are the Victorian Order of Nurses (VONs), Children's Aid Society, ParaMed, Red Cross, Rape Crisis Centres, John Howard Society, Scouts Canada, and the Girl Guides of Canada.
Part-time employment	Respondents who work for pay or in self-employment less than 30 hours per week.
Partners	Two people of the opposite sex or of the same sex who live together as a couple. These two people could be legally married to each other or living together as common-law partners. It is possible that the partners do not live in the same household (e.g. a non-cohabitant couple). In French, the word "conjoint/e" is used to designate a spouse or a common-law partner.
Peer-to-peer accommodation services	Services that connect travellers and hosts through a mobile application or website that acts as an intermediary and processes the payment from the traveller to the host.
Peer-to-peer ride services	Services that connect riders and drivers through a mobile application that acts as an intermediary and processes the payment from the rider to the driver.
Physical store	A traditional store, not online, which one can physically visit.
Platform-based peer-to-peer services	Services for sale that ultimately result in a commitment to pay, between individuals through a platform or an online marketplace. The platform acts as an intermediary or third party, by placing buyers and sellers in touch with each other. Some examples of platform-based peer-to-peer services are ride, delivery, accommodation, cleaning, handyman, plumbing or electrical work.
Podcast	A podcast is a digital audio file made available on the Internet for downloading to a computer or mobile device, typically available as a series, new installments of which can be received by subscribers automatically.
Provincial Member of Legislative Assembly	Provincial Member of Legislative Assembly, including the Premier: Provincial and territorial governments are responsible for things such as education, health care and highways. Members that are elected to a provincial government usually belong to a political party, and are called Members of Provincial Parliament (MPPs). Members that are elected to a territorial government sometimes belong to a political party, and are called Members of Legislative Assemblies (MLAs). The Premier is the head of provincial and territorial governments.
Relatives	Persons related to the respondent biologically, or by marriage, ex-marriage or common-law relationship (including all relatives such as in-laws, aunts, uncles, cousins, etc.).

Satellite broadband	Satellite broadband is transmitted wirelessly via a satellite dish, rather than relying on a network of broadband wires or cables. It works similar to satellite TV, but with satellite broadband, data can also be sent out or uploaded to the satellite. An advantage is that it can be distributed anywhere as long as there is a clear line of sight to the satellite, which means it is common in rural areas.
Separated	Persons legally married, but who no longer live with their spouse, and who have not obtained a divorce. **Note:** Couples living apart temporarily because of school, work, etc. are not considered “separated”.
Set-top box	A set-top box (traditionally located on top of your TV) decodes signals from cable, satellite or Internet service providers. Most models offer DVR/PVR and high-definition resolution display.
Single	Persons who have never married and are not living in a common law relationship.
Smartphone	A mobile phone that performs many of the functions of a computer, typically having a touchscreen interface, Internet access, and an operating system capable of running downloaded applications. Apple iPhone and Samsung Galaxy are examples of smartphones.
Smart plug	A small device that plugs into any outlet and allows users to control connected appliances wirelessly via an app.
Smart speaker	A wireless, audio playback device that uses several types of connectivity (often Wi-Fi and Bluetooth) for additional functions. Smart speakers have special features to enhance ease of use, connect to multiple types of audio sources and provide additional functionality. Some smart speakers feature digital assistants and can operate as home automation hubs. These devices are often part of a company’s existing product stack. Examples include Google Home and Amazon Echo. Alexa Voice Services (AVS), the suite of services built around Amazon's voice-controlled artificial intelligence (AI) assistant, was introduced with Echo. AVS enables voice interaction online and with various systems in the environment and online. Typically, smart speakers include Wi-Fi and Bluetooth connectivity. Simpler products range from Wi-Fi-enabled boom boxes to adjustable RGB LED smart speakers that can be spread through a home for distributed or localized stereo sound.
Smart switch	A small device that plugs into any outlet and allows users to control connected appliances wirelessly via an app.
Smart TV	A smart TV is a television set with integrated Internet capabilities or a set-top box for television that offers computing ability and Internet connectivity.
Social networking websites and apps	Social networking websites and apps host a variety of content that is shared between users with registered accounts. They are designed to allow users to share messages, photos, videos, ideas, activities, events and interests within their individual networks.

Spam	Spams are unsolicited emails, often sent with an intent to make money from its recipients through legitimate means, such as inserted advertisements, or illegitimate means, such as imbedded links that deploy viruses or gain external users digital access to your computer files.
Spouses	Two people of the opposite sex or of the same sex who live together as a couple and who are legally married to each other.
Store website associated with physical stores	The website of a retailer that is also associated with a physical store (a traditional store, not online, which one can physically visit).
Streaming	Streaming provides users with access to content, commonly audio or video, to be consumed immediately through a continuous transmission of data over the Internet, rather than through download and consumption once complete. Netflix and Crave TV are examples of streaming services that offer video content, and Spotify and Apple Music are examples of streaming services that offer audio content.
Subscriptions	Subscriptions of goods are received on a regular basis, such as bi-weekly, monthly, etc., and where payment for these goods is pre-paid at a fixed rate.
Tablet	A touch-screen monitor allowing users to navigate without the use of a keyboard (they may use their fingers or a stylus resembling a pen). It has many of the functions of a computer, the capability of connecting to the Internet, and an operating system capable of running downloaded applications. Apple iPad, Samsung Tab and Microsoft Surface are examples of tablets. t.
Used software to edit photos, video or audio files	Refers to the use of computer programs to edit photographs, video or audio files. This may include enhancing images or sound, adding special effects to videos, etc.
Used spreadsheet software advanced functions to organize and analyze data	Refers to the use of computer programs that allow for the manipulation of data in digital spreadsheets using functions that include sorting, filtering, creating charts, pivot tables and process automation with macros.
Used spreadsheet software basic functions	Used computer programs that allow for entry of text and numbers into digital spreadsheets to organize information and relate information between cells of the spreadsheet, including basic mathematic functions like adding and subtracting.
Used word processing software	Used computer programs that allow for creating, editing, saving and printing documents mainly containing text.
Virtual wallet	A virtual wallet, also called a digital wallet or an electronic wallet, is an encryption software which serves to provide the virtual equivalent of a wallet. It may contain

	digital cash, credit card information, shipping details, and a digital certificate for authentication of the wallet holder.
Virus	A computer virus is a type of malicious software program, known as malware, that, when executed, replicates itself by modifying other computer programs and inserting its own code. When this replication succeeds, the affected areas are then said to be infected with a computer virus.
Widowed	Persons who have lost their spouse through death and who have not remarried.
Website of an exclusively online retailer	The website of a retailer that operates solely online, with no association to a physical store.
Website or app selling on behalf of other businesses or individuals	The website of a retailer that operates solely online, with no association to a physical store, that sells products or services produced or provided by other companies or individuals.
Written code in a programming language	Refers to the use of computer programs to create a sequence of instructions for the computer, called code, which are written in programming language. This is done to automate performing a specific task or solving a given problem.

4.2 Demographic Concepts and Definitions

The CIUS 2022 questionnaire contains various questions to collect information about the respondent's demographic characteristics, such as location, age, gender, household size and composition, indigenous identity, population group, employment status, highest certificate, diploma or degree obtained and school attendance. The survey also collects information on the respondent's general health such as disability status and through indicators of perceived health and life satisfaction.

The CIUS 2022 database contains other demographic added through linkage with administrative files. Data on immigration status are obtained from the Longitudinal Immigration Database (IMDB) 2021, the Census family income from the Annual Estimates of Census Family Income (T1 Family File) 2021 and language data are obtained from the 2021 Census of Population.

The variables are clearly defined in the questionnaire. Additional information is included below.

Highest level of education	It is important to insist on the word “highest”. Enter one response only. Determine whether the respondents have completed their program (i.e. earned a degree, certificate or diploma) OR whether they only took some courses. If the respondent is unable to decide after some probing, choose and enter the code for the appropriate category using the information given in the questionnaire. **University certificate, diploma or degree above the bachelor's level** This category includes persons who have obtained, through a degree-granting

institution, a certificate, diploma or degree that usually requires a bachelor's degree as a prerequisite. This category includes persons who have obtained any of the following: 1) degree in medicine, dentistry, veterinary medicine or optometry; 2) master's degree or 3) earned doctorate.

Bachelor's degree
This category includes persons who have obtained a bachelor's degree awarded by a degree-granting institution and who have not obtained any higher degrees, certificates or diplomas. It includes for example, Bachelor of Arts, Bachelor of Science, and Bachelor of Laws.

University certificate or diploma below the bachelor's level
This category includes persons who have obtained a university certificate or diploma below the bachelor level and who have not obtained any higher degrees, certificates or diplomas. University certificates or diplomas are commonly connected with professional associations in fields such as accounting, banking, insurance or public administration. The certificates and diplomas referred to in this category do not require a bachelor's degree as a prerequisite.

College, CEGEP or other non-university certificate or diploma (other than trades certificates or diplomas)
This category includes persons who obtained a postsecondary certificate or diploma from a community college; a CEGEP (either general/pre-university or technical); an institute of technology; a school of nursing; a private business school; a private or public trade school; or a vocational school. Included in this category are persons who obtained a teaching or nursing certificate awarded by a provincial department of education, with the exception of teachers' or nurses' qualifications at the bachelor level or above obtained at university-affiliated faculties of education or nursing. Persons with an apprenticeship or trades certificate and no other college, CEGEP or other postsecondary and non-university certificate or diploma are excluded from this category. Persons with university certificates, diplomas or degrees are also excluded from this category.

Trades certificate or diploma
This category includes persons who have obtained a certificate, diploma or equivalent award in the skilled trades or in entry-level vocations, career and technical occupations. This may allow them to obtain or maintain an industry, government or professional designation. Persons who obtained qualifications from secondary level vocational training programs in the province of Quebec - such as the Diplôme d'études professionnelles/Diploma of Vocational Studies (DEP/DVS) - are included in this category. This category also includes persons who have obtained a journeyperson certificate in the trades through successful completion of the examinations for a Certificate of Qualification (C of Q) as well as persons with qualifications in the trades that are higher than that of pre-employment, apprenticeship or entry level. Persons who also have a postsecondary certificate, diploma or degree, other than an apprenticeship or trades certificate or diploma, are excluded from this category.

High school diploma or a high school equivalency certificate
This category includes persons who have obtained a secondary or high school diploma or a high school equivalency certificate and who have not obtained any higher certificates, diplomas or degrees. Secondary (high) school diploma or equivalency certificate includes academic or vocational high school diplomas or certificates as may be obtained by graduating from secondary school. It also includes successfully completing a high school equivalency test such as the General Educational Development (GED) test, or obtaining an Adult Basic

Education (ABE) certificate. Persons who obtained qualifications from secondary level vocational training programs in the province of Quebec - such as the Diplôme d'études professionnelles / Diploma of Vocational Studies (DEP/DVS) - are excluded from this category.
Persons who obtained any postsecondary certificate, diploma or degree are excluded from this category.
High school graduation varies from province to province and can mean anything from Grade 11 to grade 12 (formerly grade 13 in Ontario).

Province	Highest level of education
Quebec	Secondary V
Nova Scotia, Prince Edward Island, New Brunswick, Newfoundland, Alberta, British Columbia, Yukon, Northwest Territories	12

Less than high school diploma or its equivalent
This category includes persons with no schooling and persons who received some schooling but did not obtain a secondary school (high school) diploma or equivalency certificate. Persons who obtained any postsecondary certificate, diploma or degree are excluded from this category.

Interviewer note: Below are the current education system equivalents of the courses offered in the former system in Quebec:
Teacher's college: teaching certificate or one year of university
Higher seminary: Bachelor's degree in theology
Nursing school: community college diploma
Classical education: Secondary V diploma (secondary school)
Business education: trade school diploma
If the respondent completed his or her **studies in another country** and is having difficulty determining the equivalent level in the Canadian system, have them estimate the number of years of education they would have completed had they done their studies in Canada.

Immigration status

Data on immigration status are obtained through data linkage to administrative data from the Longitudinal Immigration Database (IMDB) file.

Landed immigrants are permanent residents who have indicated a year of landing in Canada since 1952.

Income

Family income data are obtained through data linkage to administrative data from the Annual Income Estimates for Census Families and Individuals (T1 Family File) T1FF tax data. Total income, in the context of census families, refers to receipts from certain sources of all of its family members, before income taxes and deductions, during a specified reference period. The monetary receipts included are those that tend to be of a regular and recurring nature.

Language

The three following linguistic questions are obtained through data linkage to Census 2021, which cover the total population. For more question, please refer to the Census data dictionary, Dictionary, Census of Population, 2021 - Complete A to Z index (statcan.gc.ca).

	Knowledge of official language Knowledge of official languages refers to whether the person can conduct a conversation in English only, French only, in both or in neither language. For a child who has not yet learned to speak, this includes languages that the child is learning to speak at home. **Language spoken most often at home** Language spoken most often at home refers to the language the person speaks most often at home at the time of data collection. A person can report more than one language as "spoken most often at home" if the languages are spoken equally often. For a person who lives alone, the language spoken most often at home is the language in which he or she feels most comfortable. For a child who has not yet learned to speak, this is the language spoken most often to the child at home. Where two languages are spoken to the child, the language spoken most often at home is the language spoken most often. If both languages are used equally often, then both languages are included here. **Mother tongue** Mother tongue refers to the first language learned at home in childhood and still understood by the person at the time the data was collected. If the person no longer understands the first language learned, the mother tongue is the second language learned. For a person who learned two languages at the same time in early childhood, the mother tongue is the language this person spoke most often at home before starting school. The person has two mother tongues only if the two languages were used equally often and are still understood by the person. For a child who has not yet learned to speak, the mother tongue is the language spoken most often to this child at home. The child has two mother tongues only if both languages are spoken
Perceived health	Perceived health refers to the perception of a person's health in general, either by the person themselves or, in the case of proxy response, by the person responding. Health means not only the absence of disease or injury but also physical, mental and social well-being.
Population group	Population group refers to whether the person reports being White, South Asian, Chinese, Black, Filipino, Arab, Latin American, Southeast Asian, West Asian, Korean, Japanese and/or another population group. These specific groups are the mark in categories used in a question which collects data on the visible minority population for employment equity purposes.

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5. Survey Methodology

The Canadian Internet Use Survey (CIUS) was administered from December 2, 2022 to April 05, 2023 Please see the following sections for descriptions of the target population, stratification, the frame, the sampling strategy, the sample size and allocation.

5.1 Target population

The target population for the 2022 CIUS included all persons 15 years of age and older in Canada, excluding:

1. Residents of the Yukon, Northwest Territories, and Nunavut; and
2. Full-time residents of institutions.

The CIUS makes efforts to identify and exclude units on reserves based on their associated geographies on the building-unit-based frame.

5.2 Stratification

In order to carry out sampling, each of the ten provinces were divided into strata (i.e. geographic areas). The stratification divides the geography of the ten provinces into rural and urban strata which were defined using the Statistical Area Classification (SAC). The three largest census metropolitan areas (CMAs), namely Montreal, Toronto and Vancouver, were separate strata.

5.3 Frame

The foundation of our survey frame was derived from the Statistical Building Register. From it, we extracted a Building Universe File. It contained geographical information and information on the type of dwelling.

The Socio-Economic File held socioeconomic and demographic information about households, including household size and composition, as well as members' ages and approximate income. It was linked to the Building Universe File to give us additional information for use when we construct survey weights.

Finally, we used the telephone file to associate telephone numbers to the buildings in our sample. Our goal was to create a frame that contained at least enough contact information for us to invite households to participate in the survey.

5.4 Sampling strategy

All records on the survey frame contained basic geographical information which allowed us to stratify the sample. Once we decided on how to allocate our sample between strata, we selected a simple random sample without replacement in each of the strata.

Once selected, the first way we tried to invite a household to participate in the survey was by sending a letter by mail. In that letter, we sent a series of instructions on who in the household was invited to participate. To be eligible, a household had to include at least one person 15 years of age or older. Of those persons 15 years of age or older, we selected a person at random. For example, in some households, we asked that the eldest member of the household participate. The person selected in the household was different depending on which letter we sent. This method of selecting a person using a variety of pre-specified instructions was called the "age-order selection method" and was a way of selecting a sample of people at random within

households. Once a person in the household was selected, we asked them to fill out the survey electronically.

If our attempt at eliciting a response was not successful, we followed up by trying to contact the household by telephone. Having made contact, we once again selected a person in the household by using the age-order selection method. If that was accomplished, the interviewer did a telephone interview to collect the data.

5.5 Sample Size and allocation by Province

The sample size for 2022 CIUS was 55,860 households. The following tables shows the allocation by province.

Province	Sample Size
Newfoundland and Labrador	3,524
Prince Edward Island	3,397
Nova Scotia	2,841
New Brunswick	2,617
Québec	11,348
Ontario	13,247
Manitoba	3,674
Saskatchewan	3,801
Alberta	4,604
British Columbia	6,807
Total	**55,860**

6. Data Collection

Data are collected through an electronic questionnaire or computer assisted telephone interviewing (CATI). No proxy reporting is allowed. The respondent has the choice to respond in French or in English. The average time to complete the survey is estimated at 30-60 minutes.

Invitation letters and reminders, containing secure access code, to complete electronic questionnaires were sent to respondents by mail. Intensive non-response follow-up was also conducted by CATI.

The collection method has changed from telephone, solely, in 2012, to electronic questionnaire in 2018, with CATI follow-up.

6.1 Non-response to the Canadian Internet Use Survey

The sample size for the 2022 CIUS was 55,860 and the number of respondents was 25,118, for a collection response rate of 45.3%.

7. Data Processing

One of the outputs of the Canadian Internet Use Survey (CIUS) is a “clean” microdata file. This chapter presents a brief summary of the processing steps involved in producing this file.

7.1 Data Capture

All responses to the 2022 CIUS questions were captured directly in the EQ application, both for the interviewer-led (iEQ) component and the respondent self-reporting (rEQ) component. The EQ application, just like any other computerized questionnaire, reduces processing time and costs associated with data entry, transcription errors and data transmission. The electronic questionnaire reduces processing time and costs associated with data entry, transcription errors and data transmission.

For some CIUS questions, data underwent a preliminary verification process when respondents were completing the survey. This was accomplished by means of a series of edits programmed into the EQ. That is, where a particular response appeared to be inconsistent with previous answers, the interviewer or the self-reporting respondent was notified with an on-screen warning message, providing them with an opportunity to modify the response provided. Once the data are collected, an extensive series of processing steps are undertaken including the editing and imputation process to identify inconsistent or missing data, and to correct errors. See Editing.

7.2 Editing

The first type of error treated was errors in questionnaire flow, where questions which did not apply to the respondent (and should therefore not have been answered) were found to contain answers. In this case a computer edit automatically eliminated superfluous data by following the flow of the questionnaire implied by answers to previous, and in some cases, subsequent questions.

The second type of error treated involved a lack of information in questions which should have been answered. For this type of error, a non-response or “not-stated” code was assigned to the item.

7.3 Imputation

Imputation is the process that supplies valid values for those variables that have been identified for a change either because of invalid information or because of missing information. The new values are supplied in such a way as to preserve the underlying structure of the data and to ensure that the resulting records will pass all required edits. In other words, the objective is not to reproduce the true microdata values, but rather to establish internally consistent data records that yield good aggregate estimates.

We can distinguish between three types of non-response. Complete non-response is when the respondent does not provide the minimum set of answers. These records are dropped and accounted for in the weighting process (see Chapter 10.0). Item non-response is when the respondent does not provide an answer to one question, but goes on to the next question. These are usually handled using the “not stated” code or are imputed. Finally, partial non-response is when the respondent provides the minimum set of answers but does not finish the interview. These records can be handled like either complete non-response or multiple item non-response.

In the case of the CIUS, donor imputation was used to fill in missing data for item and partial non-response. Further information on the imputation process is given in Chapter 8.0 (Data Quality).

7.4 Creation of Derived Variables

A number of data items on the microdata file have been derived by combining items on the questionnaire in order to facilitate data analysis. An example is the family income quintile variable, HINCQUIN which is constructed from income information. An imputation technique was used for records where the income variable was missing (see Section 8.2.3 for more details on the method used to impute income).

7.5 Weighting

The principle behind estimation in a probability sample such as CIUS is that each person in the sample "represents", besides himself or herself, several other persons not in the sample. For example, in a simple random 2% sample of the population, each person in the sample represents 50 persons in the population.

The weighting phase is a step which calculates, for each record, what this number is. This weight appears on the microdata file, and **must** be used to derive meaningful estimates from the survey. For example if the number of persons using the Internet from home is to be estimated, it is done by selecting the records referring to those individuals in the sample with that characteristic and summing the weights entered on those records.

Details of the method used to calculate these weights are presented in Chapter 10.0.

7.6 Suppression of Confidential Information

It should be noted that the "Public Use" Microdata Files (PUMF) may differ from the survey "master" files held by Statistics Canada. These differences usually are the result of actions taken to protect the anonymity of individual survey respondents. The most common actions are the suppression of data items and grouping values into wider categories. For certain variables that are susceptible to identifying individuals, the PUMF may have been treated with local suppression, that is, some of the values in the master file may have been coded as "not stated" on the PUMF.

The survey master file includes geographic identifiers for the 10 provinces as well as for the Census Metropolitan Area (CMA) and Census Agglomeration (CA), and other identifiers at the sub-provincial level. The PUMF includes geographic identifiers for the 10 provinces as well as population center indicator (CMA/CA or Outside of a CMA/CA) in every province except for Prince Edward Island.

The survey master file includes the respondent's precise age while the PUMF contains age groups only. As well, there are several other detailed variables which are only available on the survey master file.

8. Data Quality

8.1 Response Rates

The following table provides the collection response rate for the 2022 Canadian Internet Use Survey (CIUS).

Province	CIUS Respondents	CIUS Collection Response Rate (%)
Newfoundland and Labrador	1,191	34.0
Prince Edward Island	1,064	31.6
Nova Scotia	1,308	46.1
New Brunswick	1,013	38.8
Québec	5,665	50.0
Ontario	6,462	48.9
Manitoba	1,496	40.6
Saskatchewan	1,532	40.2
Alberta	2,016	43.7
British Columbia	3,371	49.4
Canada	25,118	45.3

8.2 Survey Errors

The estimates derived from this survey are based on a sample of households. Somewhat different estimates might have been obtained if a complete census had been taken using the same questionnaire, interviewers, supervisors, processing methods, etc. as those actually used in the survey. The difference between the estimates obtained from the sample and those resulting from a complete count taken under similar conditions, is called the sampling error of the estimate.

Errors which are not related to sampling may occur at almost every phase of a survey operation. Interviewers may misunderstand instructions, respondents may make errors in answering questions, the answers may be incorrectly entered on the questionnaire and errors may be introduced in the processing and tabulation of the data. These are all examples of non-sampling errors.

Over a large number of observations, randomly occurring errors will have little effect on estimates derived from the survey. However, errors occurring systematically will contribute to biases in the survey estimates. Considerable time and effort were taken to reduce non-sampling errors in the survey. Quality assurance measures were implemented at each step of the data collection and processing cycle to monitor the quality of the data. These measures include the use of highly skilled interviewers, extensive training of interviewers with respect to the survey procedures and questionnaire, observation of interviewers to detect problems of questionnaire design or misunderstanding of instructions, procedures to ensure that data capture errors were minimized, and coding and edit quality checks to verify the processing logic.

8.2.1 Data Collection

Interviewer training consisted of reading the CIUS Interviewer's Manual, practicing with the CIUS training cases on the computer and discussing any questions with senior interviewers before the start of the survey. A description of the background and objectives of the survey was provided, as well as a glossary of terms and a set of

questions and answers. The collection period ran from December 02, 2022 to April 05, 2023.

8.2.2 Data Processing

Data processing of the CIUS was done in a number of steps including verification, coding, editing, imputation, estimation, confidentiality, etc. At each step a picture of the output files is taken and an easy verification can be made comparing files at the current and previous step. This greatly improved the data processing stage.

8.2.3 Non-response

A major source of non-sampling errors in surveys is the effect of non-response on the survey results. The extent of non-response varies from partial non-response (failure to answer just one or some questions) to total non-response. Total non-response occurred because the interviewer was either unable to contact the respondent, no member of the household was able to provide the information, or the respondent refused to participate in the survey. Total non-response was handled by adjusting the weight of individuals who responded to the survey to compensate for those who did not respond.

In most cases, partial non-response to the survey occurred when the respondent did not understand or misinterpreted a question, refused to answer a question, or could not recall the requested information. In order to provide complete data concerning the distribution of family income among sampled units and concerning the calculation of totals for expenditures of physical goods and services ordered over the Internet, values were imputed when these were missing.

The imputations involved donors that were selected using a score function. For each item non-response or partial non-response records (also called recipient records), certain characteristics were compared to characteristics from all the donors. When the characteristics were the same between a donor and the recipient, a value was added to the score of that donor. The donor with the highest score was deemed the “closest” donor and was chosen to fill in missing pieces of information of the non-respondent. If there was more than one donor with the highest score, a random selection occurred. The pool of donors was made up in such a way that the imputed value assigned to the recipient, in conjunction with other non-imputed items from the recipient, would still pass the edits.

The CIUS imputation process helped to fill incomplete responses with the experience of other respondents with similar or identical characteristics. This will add to the number of units used in any analysis performed by researchers.

8.2.4 Measurement of sampling error

Since it is an unavoidable fact that estimates from a sample survey are subject to sampling error, sound statistical practice calls for researchers to provide users with some indication of the magnitude of this sampling error. This section of the documentation outlines the measures of sampling error which Statistics Canada commonly uses and which it urges users producing estimates from this microdata file to use also.

The basis for measuring the potential size of sampling errors is the standard error of the estimates derived from survey results.

However, because of the large variety of estimates that can be produced from a survey, the standard error of an estimate is usually expressed relative to the estimate to which it pertains. This resulting measure, known as the coefficient of variation (CV) of an

estimate, is obtained by dividing the standard error of the estimate by the estimate itself and is expressed as a percentage of the estimate.

There is more information on the calculation of coefficients of variation in Appendix A.

9.0 Guidelines for Tabulation, Analysis and Release

This chapter of the documentation outlines the guidelines to be adhered to by users tabulating, analyzing, publishing or otherwise releasing any data derived from the survey microdata files. With the aid of these guidelines, users of microdata should be able to produce the same figures as those produced by Statistics Canada and, at the same time, will be able to develop currently unpublished figures in a manner consistent with these established guidelines.

9.1 Rounding Guidelines

In order that estimates for publication or other release derived from these microdata files correspond to those produced by Statistics Canada, users are urged to adhere to the following guidelines regarding the rounding of such estimates:

a) Estimates in the main body of a statistical table are to be rounded to the nearest hundred units using the normal rounding technique. In normal rounding, if the first or only digit to be dropped is 0 to 4, the last digit to be retained is not changed. If the first or only digit to be dropped is 5 to 9, the last digit to be retained is raised by one. For example, in normal rounding to the nearest 100, if the last two digits are between 00 and 49, they are changed to 00 and the preceding digit (the hundreds digit) is left unchanged. If the last digits are between 50 and 99 they are changed to 00 and the preceding digit is incremented by 1.

b) Marginal sub-totals and totals in statistical tables are to be derived from their corresponding unrounded components and then are to be rounded themselves to the nearest 100 units using normal rounding.

c) Averages, proportions, rates and percentages are to be computed from unrounded components (i.e. numerators and/or denominators) and then are to be rounded themselves to one decimal using normal rounding. In normal rounding to a single digit, if the final or only digit to be dropped is 0 to 4, the last digit to be retained is not changed. If the first or only digit to be dropped is 5 to 9, the last digit to be retained is increased by 1.

d) Sums and differences of aggregates (or ratio) are to be derived from their corresponding unrounded components and then are to be rounded themselves to the nearest 100 units (or the nearest one decimal) using normal rounding.

e) In instances where, due to technical or other limitations, a rounding technique other than normal rounding is used resulting in estimates to be published or otherwise released which differ from corresponding estimates published by Statistics Canada, users are urged to note the reason for such differences in the publication or release document(s).

f) Under no circumstances are unrounded estimates to be published or otherwise released by users. Unrounded estimates imply greater precision than actually exists.

9.2 Sample Weighting Guidelines for Tabulation

The sample design used for the Canadian Internet Use Survey (CIUS) was not self-weighting. When producing simple estimates, including the production of ordinary statistical tables, users must apply the proper sampling weight.

If proper weights are not used, the estimates derived from the microdata files cannot be considered to be representative of the survey population, and will not correspond to those produced by Statistics Canada.

Users should also note that some software packages may not allow the generation of estimates that exactly match those available from Statistics Canada, because of their treatment of the weight field.

9.3 Definitions of Types of Estimates: Categorical and Quantitative

Before discussing how the CIUS data can be tabulated and analyzed, it is useful to describe the two main types of point estimates of population characteristics which can be generated from the microdata file for the CIUS.

9.3.1 Categorical Estimates

Categorical estimates are estimates of the number, or percentage of the surveyed population possessing certain characteristics or falling into some defined category. The number of Canadians who did not use the Internet during the past three months or the proportion of Canadians who watched 20 hours or more per week of video streaming content, in a typical week are examples of such estimates. An estimate of the number of persons possessing a certain characteristic may also be referred to as an estimate of an aggregate.

Examples of Categorical Questions:

Q: During the past three months, have you used the Internet for personal use?
R: Yes / No

Q: During a typical week, how many hours do you spend on the following activities over the Internet? a. Watching streamed content
R: None / Less than 10 hours per week 10 to less than 20 hours per week / 20 hours or more per week

9.3.2 Quantitative Estimates

Quantitative estimates are estimates of totals or of means, medians and other measures of central tendency of quantities based upon some or all of the members of the surveyed population. They also specifically involve estimates of the form $\hat{X}/\hat{Y}$ where $\hat{X}$ is an estimate of surveyed population quantity total and $\hat{Y}$ is an estimate of the number of persons in the surveyed population contributing to that total quantity.

An example of a quantitative estimate is the average amount spent on physical goods by Canadians during the past 12 months over the Internet. The numerator is an estimate of the total amount spent on physical goods and its denominator is the number of persons spending at least $ 1 on physical good.

Examples of Quantitative Questions:

Q: How much did you spend online on physical goods during the past 12 months?
R: |_|_|_| Amount rounded to the nearest CAN$

9.3.3 Tabulation of Categorical Estimates

Estimates of the number of persons with a certain characteristic can be obtained from the microdata file by summing the final weights of all records possessing the characteristic(s) of interest. Proportions and ratios of the form $\hat{X}/\hat{Y}$ are obtained by:

a) summing the final weights of records having the characteristic of interest for the numerator ($\hat{X}$),
b) summing the final weights of records having the characteristic of interest for the denominator ($\hat{Y}$), then
c) dividing estimate a) by estimate b) ($\hat{X}/\hat{Y}$).

9.3.4 Tabulation of Quantitative Estimates

Estimates of quantities can be obtained from the microdata file by multiplying the value of the variable of interest by the final weight for each record, then summing this quantity over all records of interest. For example, to obtain an estimate of total spending on physical goods by Canadian males during the past 12 months over the Internet, multiply the value reported in question EC_0120A (amounts spent on physical goods) by the final weight for the record, then sum this value over all records with GENDER = 1 (males).

To obtain a weighted average of the form $\hat{X}/\hat{Y}$, the numerator ($\hat{X}$) is calculated as for a quantitative estimate and the denominator ($\hat{Y}$) is calculated as for a categorical estimate. For example, to estimate the average spending for physical goods made by Canadian males during the past 12 months over the Internet,

a) estimate the total amount spent ($\hat{X}$) as described above,
b) estimate the number of persons ($\hat{Y}$) in this category by summing the final weights of all records with GENDER = 1, then
c) divide estimate a) by estimate b) ($\hat{X}/\hat{Y}$).

9.4 Guidelines for Statistical Analysis

The CIUS sample design stratifies rural and urban households within provinces and then randomly selects individuals within a household, resulting in unequal probabilities of selection of respondents. Using data from such complex surveys presents a challenge to analysts because the survey design and uneven non-response patterns affect the survey weights which are calibrated to reliable provincial demographic projections. To mitigate potential bias in survey estimates, the survey weight (variable name WTPG) must be used when using the CIUS data files.
While many analysis procedures found in statistical packages allow weights to be used, the meaning or definition of the weight in these procedures may differ from that which is appropriate in a sample survey framework, with the result that while in many cases the estimates produced by the packages are correct, the variances that are calculated are poor.

For many analysis techniques (for example linear regression, logistic regression and analysis of variance), a method exists which can make the variances calculated by the standard packages more meaningful, by incorporating the unequal probabilities of selection. The method rescales the weights so that there is an average weight of 1.

For example, suppose that analysis of all male respondents is required. The steps to rescale the weights are as follows:

1) select all respondents from the file who reported GENDER = 1, men;
2) calculate the AVERAGE weight for these records by summing the original person weights from the microdata file for these records and then dividing by the number of respondents who reported GENDER = 1;
3) for each of these respondents, calculate a RESCALED weight equal to the original person weight divided by the AVERAGE weight;

4) perform the analysis for these respondents using the RESCALED weight.

The variances produced by the standard packages using the RESCALED weight will be more reasonable; they still will not take into account the stratification and clustering of the sample's design, but they will take into account the unequal probabilities of selection.

9.5 Coefficient of Variation Release Guidelines

Before releasing and/or publishing any estimates from the CIUS, users should first determine the quality level of the estimate. The quality levels are acceptable, marginal and unacceptable. Data quality is affected by both sampling and non-sampling errors as discussed in Chapter 8.0. However for this purpose, the quality level of an estimate will be determined only on the basis of sampling error as reflected by the coefficient of variation as shown in the table below. Nonetheless users should be sure to read Chapter 8.0 to be more fully aware of the quality characteristics of these data.

First, the number of respondents who contribute to the calculation of the estimate, the denominator, should be determined. If this number is less than 15, the weighted estimate should be considered to be of unacceptable quality.

For weighted estimates based on sample sizes of 15 or more, users should determine the coefficient of variation of the estimate and follow the guidelines below. These quality level guidelines should be applied to rounded weighted estimates.
All estimates can be considered releasable. However, those of marginal or unacceptable quality level must be accompanied by a warning to caution subsequent users.

Quality Level Guidelines

When considering releasing *and/or* publishing an estimate from the analytical file, users should consult the table below and follow the guideline that matches the coefficient of variation of the estimate.

Quality Level of Estimate	Guidelines
1) Acceptable	Estimates have a sample size of 15 or more, and low coefficients of variation in the range of 0.0% to 16.5%. No warning is required.
2) Marginal	Estimates have a sample size of 15 or more, and high coefficients of variation in the range of 16.6% to 33.3%. Estimates should be flagged with the letter E (or some similar identifier). They should be accompanied by a warning to caution subsequent users about the high levels of error, associated with the estimates.
3) Unacceptable	Estimates have a sample size of less than 15, or very high coefficients of variation in excess of 33.3%. Statistics Canada recommends not to release estimates of unacceptable quality. However, if the user chooses to do so then estimates should be flagged with the letter F (or some similar identifier) and the following warning should accompany the estimates: “Please be warned that these estimates [flagged with the letter F] do not meet Statistics Canada’s quality standards. Conclusions based on these data will be unreliable, and most likely invalid.”

10. Weighting

The weighting factor WTPG was placed on the Main File. This is the weighting factor for analysis at the person level, i.e. to calculate estimates of the number of persons (non-institutionalized and aged 15 or over) having one or several given characteristics. WTPG should be used for all person-level estimates. For example, to estimate the number of persons who reported that their use of the Internet in a typical week is 20 hours or more, the value of WTPG is summed over all records with this characteristic.

In addition to the estimation weight, bootstrap weights have been created for the purpose of design-based variance estimation[1].

10.1 Weighting Procedures

The records on the survey frame are households. A simple random sample of those records was selected in each stratum. Therefore, each record within a stratum has an equal probability of selection.

a) Initial weight calculation

The sample design was a two-stage sample design. The first stage was a sample of households. Knowing the number of households on the frame by stratum, it was possible to calculate initial probabilities of selection for the initial sample.

b) Removal of out-of-scope records

Some of the records on the frame should not have been there because they were either vacant or were businesses, institutions, or other out-of-scope dwellings. Out-of-scope records were simply removed from the process, leaving only in-scope records in the sample.

c) Non-response adjustment

The non-response adjustment was done at the household level. Estimated total household income category by region (Eastern, Quebec, Ontario and Western) formed the non-response weighting classes.

d) Person weight calculation

For this last step, we adjusted the household weight to reflect the people who responded to the survey. Using age group and sex, the household weights were modified so that the sum of the weights equated to reliable totals. These totals were demographic projections of the 15+ population which were based, in part, on the 2021 Census of Population. This calibration changed the household level weight in the previous step to a person level weight. Both the geography of the household and the age and sex of the person who responded were used to calibrate the weight. This calculated weight is the final person weight (WTPG) which appears on the CIUS person master file as well as on the person public use microdata file.

10.2 Weighting Policy

Users are cautioned against releasing unweighted tables or performing any analysis based on unweighted survey results. As was discussed in Section 10.1, several weight adjustments were performed that depended on the province, age and sex of the respondent. Sampling rates as well as non-response rates varied significantly from province to province, and non-response rates varied with demographic characteristics. For example, non-respondents are often more likely to

[1] 1,000 standard bootstrap weights are available for the 2022 CIUS: WRPG1-WRPG1000 for the 25,118 respondents.

be males and more likely to be younger. In the responding sample, 2.4% were males between the ages of 15- and 24, while in the overall population, approximately 7.4% were males between 15 and 24. Therefore, it is clear that unweighted sample counts cannot be considered representative of the survey target population.

11. Survey specific notes

11.1 E-commerce

There was a methodological change for the collection of the e-commerce indicators in 2022, including a revision in question structure. Use caution when making comparisons between cycles.

12. Questionnaire

12.1 The Canadian Internet Use Survey Questionnaire

The Canadian Internet Use Survey (CIUS) questionnaire is available online here: Canadian Internet Use Survey 2022 (statcan.gc.ca)

Appendix A – Variance Estimation for Master and Share Files

In order to determine the quality of the estimate and to calculate the coefficient of variation (CV), the standard deviation must be calculated. Confidence intervals also require the standard deviation of the estimate. The Canadian Internet Use Survey (CIUS) uses a multi-stage survey design and calibration, which means that there is no simple formula that can be used to calculate variance estimates. The survey therefore uses a resampling method called the bootstrap.

Bootstrap Variance Estimation

The Bootstrap method belongs to a family of replicate-based variance estimation techniques. A detailed discussion of replication methods can be found in Wolter (2007), Rust & Rao (1996) or Lohr (1999). Such methods involve the taking of repeated subsamples, or replicates, from the data, re-computing the weighted survey estimate for each replicate, and the full sample, and then computing the variance as a function of the resulting estimates. Other replication methods include Jackknife and Balanced Repeated Replication (BRR).

Several Statistics Canada surveys rely on the bootstrap method to estimate the sampling variance. The method is accessible through a survey-specific set of bootstrap weights that are often provided directly with the survey microdata.

Software Options

In order to properly estimate the variance of a given estimate using replication methods, several different commercially available software packages are available. Examples include SAS, Stata, SUDAAN, and WesVar. While the bootstrap method is not often directly supported, the BRR is supported and the functionality can be used to estimate the variance for surveys using the bootstrap method. The bootstrap and BRR differ in the way in which the replicates are built but the variance estimation process is the same for both methods. This is a point that some software documentation fails to mention in great detail (Phillips, 2004).

More information on these software can be found on the following websites:

SAS: http://www.sas.com

SAS 9.2 is the first version of SAS that offers some replication approaches to variance estimation in its four survey analysis procedures. The BRR option can be used with user-supplied bootstrap weights in order to obtain bootstrap variance estimates (Phillips, 2004).

Stata: http://www.stata.com

Stata 12 offers a full suite of design-based variance estimation options with the svy commands – Taylor series, jackknife, BRR, and bootstrap. The bootstrap option can be used with user-specified survey bootstrap weights, such as those provided with many Statistics Canada surveys, in order to obtain bootstrap variance estimates.

SUDAAN: http://www.rti.org/sudaan

SUDAAN 10 offers three variance estimation methods – Taylor series, jackknife and BRR. The BRR option can be used with user-specified bootstrap weights in order to obtain bootstrap variance estimates (Phillips, 2004).

WesVar: http://www.westat.com

WesVar carries out various analyses of survey data using replication methods for variance estimation. One of the methods offered is BRR with a Fay adjustment, which, as explained in Phillips (2004), can be used to get bootstrap variance estimates if the bootstrap weight variables are provided by the researcher.

Additionally, any software that offers an analytic procedure or command that can produce weighted estimates of the parameters of interest and also has the flexibility of a programming language may be used iteratively to obtain bootstrap variance estimates.

Bootstrap in Practice

A detailed description (along with examples) of how to implement the bootstrap method with the selected software is found in Statistics Canada's Research Data Centre's Information and Technical Bulletin – "Weighted estimation and bootstrap variance estimation for analyzing survey data: How to implement in selected software" (Gagné, Roberts et Keown, 2014, 2014).
http://www.statcan.gc.ca/pub/12-002-x/2014001/article/11901-eng.htm

With each software option for estimating the bootstrap variance, certain key files, variables and parameters are required. The information required includes:

Files[2]:
The data file – The file containing all observations and their characteristics collected by the survey.
The bootstrap weight file – The file containing the weights for each of the bootstrap replicates.

Variables:

- **Identification variable(s) –** Unique identifiers for every observation on the data file
- **The survey weight** – The weight to be used in calculating the survey estimates.
- **The replicate or bootstrap weights –**There is one weight for each bootstrap replicate
- **Analysis variables –** These are variables to be used by the researcher in their analysis. These may be directly available on the data files or derived for the purposes of the study.

Parameters:

- The replicate weights' names or prefix (Ex. WRPG1-WRPG1000)
- The number of replicate weights specific to the survey. For CIUS, there are 1,000 bootstrap replicates.
- A mean bootstrap parameter (Fay adjustment or Fay's Factor) if the survey used the mean bootstrap methodology. CIUS did not use the mean bootstrap methodology.

[2] Most software packages require that the 2 files be merged into 1 analytical file. This can be completed by the user via the unique identification variables present in both files.

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